

Nexo Medico AI - System Modules & Architecture Guide

Complete Technical Specifications, Role Portals, Workflows & AI Engines

1. System Executive Overview & Tech Stack

Nexo Medico AI is an enterprise-grade multi-tenant hospital management platform integrating clinical inpatient/outpatient workflows with specialized real-time Clinical & Administrative AI Engines.

Key Technology Stack:

- Framework: Next.js 14 App Router (React 18, TypeScript, Tailwind CSS, Lucide Icons)
- Database & ORM: PostgreSQL / SQLite via Prisma ORM (55+ relational database models)
- Authentication: JWT Authentication with Role-Based Access Control (RBAC) & hospitalId tenant scoping
- AI & Speech: Web Speech API (Speech-to-Text), Dual-Mode TTS Audio Synthesis, BioBERT Vector RAG

2. Multi-Tenant Role Portals Summary

- Platform Super Admin (SUPER_ADMIN) [Route: /admin/dashboard]

Universal platform governance, registered hospitals directory, global user directory, support ticket management center, global operations control center, and AI platform security suite.

- Hospital Tenant Admin (HOSPITAL_ADMIN) [Route: /admin/dashboard]

Hospital settings, department creation, physician & nursing staff allocations, inpatient bed management, pharmacy & laboratory oversight.

- Physician / Doctor (DOCTOR) [Route: /doctor/dashboard]

Patient consultation schedule (/doctor/appointments), EHR medical timeline, digital prescription generator, radiology workstation.

- Nurse (NURSE / STAFF) [Route: /nurse/beds]

Inpatient ward & bed grid (Occupied/Available/Cleaning beds), vital sign entry, ICU monitoring logs, patient admission & discharge.

- Pharmacist (PHARMACIST) [Route: /pharmacy/dashboard]

Prescriptions queue, medicine inventory stock management, drug-drug interaction safety checker.

- Lab Technician (LAB_TECH) [Route: /laboratory/dashboard]

Diagnostic test orders queue, OCR report text scanner, test result validation.

- Patient (PATIENT) [Route: /patient/dashboard]

Personal health dashboard, 7 Specialized AI Clinical & Voice Engines, medical timeline, prescriptions, lab reports, longevity score.

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3. Detailed Platform Modules & Functionality

Module A: Platform Master Governance & Super Admin

- Registered Hospitals Directory (/admin/hospitals): Onboard, edit, and suspend hospital tenants.
- Universal Patient Registry (/admin/patients): Universal patient registry across all platform tenants.
- Support & Issue Tickets Center (/admin/support-issues): Live management of user-reported issues.
- Global Operations Control Center (/admin/operations): 4 workstream tabs (Appointments, Admissions, Prescriptions, Labs).
- Platform Security & Audit Logs (/admin/security, /admin/audit-logs): Live security audit trail.

Module B: Clinical Doctor & Physician Portal

- Patient Appointments Schedule (/doctor/appointments): Daily assigned patient consultation schedule with time slots, patient code, and chief complaints.
- Medical EHR Timeline (/doctor/patients): Chronological patient records, diagnostic history, and doctor notes.
- Prescription Engine (/doctor/prescriptions): Digital prescription authoring with dosage, frequency, and duration.
- Radiology Workstation (/doctor/radiology): Medical image viewing and X-ray diagnosis.

Module C: Nurse Wards, Bed Grid & ICU Center

- Inpatient Wards & Bed Grid (/nurse/beds): Visual color-coded bed status cards (Occupied - Red, Available - Green, Cleaning - Amber) with interactive patient admission & discharge modals.
- ICU Monitoring Logs (/nurse/icu-logs): Vital sign logs (BP, HR, SpO2, Temp) and trend graphs.

Module D: Pharmacy & Inventory Management

- Prescription Dispensing Queue (/pharmacy/dashboard): Live fulfillment queue for patient prescriptions.
- Medicine Inventory (/pharmacy/inventory): Stock tracking, reorder alerts, and expiry management.
- Drug Interaction Safety Checker (/pharmacy/drug-checker): Multi-drug contraindication analysis.

Module E: Laboratory & OCR Diagnostics

- Test Orders Queue (/laboratory/dashboard): Lab sample processing and result verification.
- OCR Report Upload (/laboratory/ocr-upload): Optical Character Recognition scanning to convert paper lab reports into structured digital records.

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4. Specialized AI Supercomputer Engines Suite

A. Patient & Clinical AI Engines (/patient/ai-engines)

1. Multilingual Voice AI Recognition & Diagnostic Triage

Real-time Speech-to-Text dictation & Dual-Mode Text-to-Speech audio in 9+ languages (English, Telugu, Hindi, Tamil, Spanish, French, German, Mandarin, Arabic). Analyzes queries in real time and plays audio responses.

2. BioBERT RAG Symptom Triage

Retrieves vector medical knowledge with PubMed citations for differential diagnosis & OTC guidance.

3. Lab OCR Report Simplifier

Parses complex lab test values (HbA1c, Glucose, Lipids, WBC) into plain-text interpretations.

4. Polypharmacy Drug Interaction Checker

Evaluates multi-drug combinations for toxic interactions and contraindications.

5. Chronic Cardiovascular Risk Engine

Calculates 10-year Framingham/ASCVD risk scores using BP, BMI, glucose, and lifestyle metrics.

6. Mental Wellness & Sleep Protocol

Stress score calculator and cognitive sleep hygiene protocol generator.

7. Clinical Nutrition Plan Generator

Generates disease-specific 7-day meal protocols for diabetes, hypertension, and kidney health.

B. Super Admin Infrastructure AI Engines (/admin/ai-engines)

1. Cyber Threat Anomaly Detector

Real-time DDoS, brute-force intrusion, and IP anomaly score detector.

2. Hospital Compliance & HIPAA Auditor

Automated multi-tenant regulatory risk and compliance auditor.

3. Healthcare Revenue Predictor

Predictive revenue forecasting based on bed occupancy and claim approval rates.

4. System Server Health Monitor

Real-time server latency, DB connection pool, and Redis cache inspector.

5. AI Model Governance Auditor

Clinical model drift tracker, hallucination index calculator, and audit logger.

6. Support Ticket Auto-Triage NLP

Sentiment analysis and priority classification for hospital support issues.

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5. Global Floating Nexo AI Assistant & Support Engine

The Floating AI Bot (FloatingAiChatbot.tsx) is rendered globally across all pages:

- Voice Recognition Dictation: Microphone button allows dictation anywhere.
- Audio Voice Response Toggle: Toggle speech audio playback for AI replies.
- Support Ticket Creation: Interactive ticket form capturing Contact Name, Mobile, Email, Category, Subject, Description, and Screenshot Evidence Image attachment.
- Live Synchronization: Syncs tickets directly to /admin/support-issues generating TICKET-2026-XXXX.

6. Database Schema Overview (Prisma Models)

The relational database (dev.db) consists of 55+ entities including:

- Hospital, Department, User, Role, Permission, RolePermission, UserPermission
- StaffProfile, DoctorProfile, NurseProfile, Patient
- MedicalRecord, Appointment, Consultation, Prescription, Medicine, PharmacyInventory, DispensingLog
- LabTest, LabOrder, LabReport, MedicalImage
- Ward, Bed, Admission, Discharge, ICUUnit, ICUBedLog
- AIPredictionLog, SupportIssue, AuditLog